

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 2

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Find the term that is independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$. [3]

2 Find the set of values of the constant k for which the line $y = kx - 3$ does not meet the curve $y = x^2 + 2x + 1$. [4]

3 A sector of a circle has radius r cm and angle θ radians. The perimeter of the sector is 20 cm and its area is 16 cm^2 . Find the values of r and θ , justifying your rejection of any solution that is not valid. [4]

4 The points A , B and C have coordinates $(-2, 1)$, $(4, 5)$ and $(6, 2)$ respectively.

(a) Show that angle ABC is a right angle. [2]

(b) Find the coordinates of the point D such that $ABCD$ is a rectangle. [1]

(c) Find the area of the rectangle $ABCD$. [2]

5

(a) Prove the identity $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \equiv \frac{2}{\sin \theta}$. [3]

(b) Hence solve the equation $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 3$ for $0 < \theta < 2\pi$, giving your answers in radians correct to 3 significant figures. [3]

6 The function f is defined by $f(x) = 4 - x^2$ for $x \leq 0$.

(a) State the range of f . [1]

(b) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [3]

(c) The function g is defined by $g(x) = 2x + 1$ for $x \in \mathbb{R}$. Solve the equation $gf(x) = -9$. [3]

7 A curve has equation $y = (2x - 3)^3 - 6x$.

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [3]

(b) Find the coordinates of the stationary points of the curve and determine the nature of each stationary point. [4]

8

(a) The volume of a spherical balloon is increasing at a constant rate of $12 \text{ cm}^3 \text{ s}^{-1}$. Find the rate of increase of the radius at the instant when the radius is 2 cm. [The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.] [3]

(b) Find the coordinates of the point on the curve $y = x^2 - 3x + 5$ at which the tangent is parallel to the line $y = 5x - 1$, and find the equation of this tangent. [4]

9

(a) The sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 2n$. Find the first term and the common difference of the progression, and find the 20th term. [4]

(b) The first three terms of a geometric progression are $(x + 4)$, $(x - 2)$ and $(x - 5)$.

(i) Find the value of x . [2]

(ii) Find the sum to infinity of the progression. [2]

10 The curve $y = 9 - x^2$ meets the line $y = 5$ at the points A and B .

(a) Find the area of the region enclosed by the curve and the line AB . **[4]**

(b) The region bounded by the curve $y = 9 - x^2$, the positive x -axis and the positive y -axis is rotated through 360° about the y -axis. Find the volume of the solid formed, giving your answer in terms of π . **[4]**

11 A circle has centre $C(2, -1)$ and passes through the point $(5, 3)$.

(a) Find the equation of the circle. **[2]**

(b) The line $x + 2y = 10$ meets the circle at the points P and Q . Find the coordinates of P and Q . **[3]**

(c) Find the equation of the tangent to the circle at the point with coordinates $(6, 2)$, giving your answer in the form $ax + by = c$. **[4]**

12

(a) Solve the equation $x - 5\sqrt{x} + 6 = 0$. **[3]**

(b) Find the set of values of the constant k for which the equation $kx^2 + 4x + k = 0$ has two distinct real roots. **[4]**

END OF PAPER (Total: 75 marks)